

THE RHYTHMS OF RELIEF: DESIGNING A STRESS-RELIEVING MUSIC RECOMMENDER SYSTEM FOR STUDENTS

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ABSTRACT

Stress is a common challenge for university students, and music offers a potential coping mechanism. This study develops a prototype for a hybrid music recommender system that personalizes stress-relief music based on user preferences and audio features. A survey collected student-curated stress-relief songs, which were clustered using k-Means to identify relaxation-focused and energy-driven coping strategies. A prototype was tested by 17 students to evaluate its effectiveness and usability. Results showed strong acceptance towards the prototype, but highlighted the need for improvements in scalability, recommendation adaptability, and feature extraction techniques. Overall, the study highlights the potential of hybrid recommender systems in supporting student stress management.

1. INTRODUCTION AND RELATED WORKS

1.1 What is stress and why should we mitigate it?

Stress is broadly defined as the “perceive[d] discrepancy between the demands of a situation and the resources available to meet those demands” [1]. It can manifest itself through both physical and physiological dimensions, and is known to significantly impact mental and physical health [2]. Physiologically, stress triggers biological responses such as increased heart rate, elevated cortisol levels, and immune system suppression, which can contribute to long-term health complications [2]. Psychologically, stress is shaped by an individual’s subjective perception of environmental demands and their perceived ability to cope with those demands [3].

Dougall and Baum [3] conceptualise stress as a multifaceted phenomenon, emphasising its physical and psychological complexity through three dimensions. Firstly, stress can be seen as an *environment*, in which certain external events or conditions can act as triggers (or, *stressors*). Secondly, it can be seen as a *response*, focusing on one’s physical and emotional reactions to these stressors. Lastly, stress can be understood as a *process*,

highlighting the dynamic interaction between external stressors and an individual’s cognitive, emotional, and behavioral coping mechanisms. This multidimensional approach underscores the dynamic relationship between humans and their stressful surroundings, wherein stress is not merely a static condition, but an evolving interaction with one’s environment.

Chronic stress, if left unmanaged, can have profound consequences for both physical and mental health. Prolonged exposure to stress has been linked to cardiovascular diseases, immune dysfunction, and increasing susceptibility to physical illness [4]. Additionally, it is strongly associated with mental health disorders, including anxiety, burnout, and depression [2]. Given these far-reaching implications, mitigating stress is crucial for enhancing overall well-being and preventing long-term physical and psychological harm.

1.2 Stress among students

Academic-related stress is a particularly pervasive challenge among students, with common concerns including stress about grades, tests, and study performance. Research from Pascoe et al. [5] indicates that 66% of students report concerns about poor grades, 55% feel anxious over exams even when well-prepared, and 37% experience stressful tension while studying. Through its frequency and persistence, academic stress can have a significant impact on students’ physical and mental health, contributing to fatigue, anxiety, and depression [2].

Shankar and Park [2] propose a conceptual model illustrating how stress impacts students’ health and academic performance. They argue that stress exerts both direct and indirect effects: physiologically, it places strain on the body, leading to adverse health outcomes, while behaviorally, it can result in unhealthy coping mechanisms such as sleep deprivation and substance use [4–6]. Additionally, stress can reduce academic motivation, lower engagement levels, and increase likelihood of dropout.

Because of these direct impacts that stress can have, effective stress management and healthy coping strategies are vital in improving student’s well-being and academic success. Lazarus and Folkman [7] categorise coping mechanisms into two primary categories: *problem-focused* coping and *emotion-focused* coping. Problem-focused coping proactive measures to address the root cause of stress,



such as improving time management, restructuring study habits, or seeking academic assistance. Emotion-focused coping, in contrast, aims to regulate the emotional response to stress through methods such as mindfulness, relaxation techniques, and seeking social support.

Providing and encouraging healthy strategies of both problem-focused and emotion-focused coping to students can decrease the physical and psychological consequences of stress [2]. This could also help students to avoid unhealthy coping mechanisms such as substance abuse [2, 4–6]. One promising emotion-focused approach is the use of music as a tool for stress reduction.

1.3 Music and stress reduction

Music has been recognised for its potential to reduce stress, with evidence suggesting that it can reduce both physical stress indicators (f.e. heart rate and blood pressure) and psychological symptoms (f.e. anxiety and nervousness) [4, 8]. Music can function as an emotion-focused coping mechanism by redirecting attention away from the negative stimuli and activating brain regions associated with memory, emotion and mood [8, 9]. Due to its adaptability and effectiveness, music therapy has been successfully implemented in the treatment and support for individuals with autism, ADHD, and cancer [10].

1.3.1 Musical genres and stress reduction

Extensive research has explored the relationship between musical genres and stress reduction. Certain genres, such as classical and jazz, have been found to promote relaxation and lower physiological stress markers. For instance, classical music has been shown to decrease heart rate and perceived anxiety [11], while jazz music has been linked to reduced cortisol levels - the primary stress hormone - suggesting a calming effect [12]. Conversely, some genres may induce or exacerbate stress rather than alleviate it. Heavy metal, for example, has been associated with heightened anxiety in some listeners. Labbé [13] found that participants experienced increased levels of state anxiety after exposure to heavy metal music. Similarly, techno music has been shown to elevate cortisol levels, particularly in individuals prone to anxiety, suggesting that its stimulating effects may contribute to stress rather than mitigate it [14]. Additionally, exposure to violent and aggressive song lyrics has been linked to increased stress and anxiety, particularly in listeners already susceptible to stress-related disorders [15].

Instrumental music appears to be more effective than lyrical music in reducing physiological stress [16]. This is likely because instrumental tracks promote relaxation without engaging cognitive processes tied to language, allowing the listener to disengage from stressful thoughts more effectively [4].

Although research on musical genres provides valuable insights, genres themselves are broad and subjective, varying based on cultural background, personal experience, and individual preferences. Therefore, rather than focusing solely on genre classifications, a more precise approach is

to examine specific *musical features* - such as tempo, frequency, and instrumentation - that contribute to stress reduction. Identifying the most effective characteristics for relaxation could lead to a more tailored and personalised understanding of music's stress-relieving potential.

1.3.2 Musical features and stress reduction

Musical features, such as tempo and frequency, can play a crucial role in stress reduction. High-frequency music, particularly in the range of 528 Hz, has been found to lower cortisol levels and increase oxytocin - a hormone associated with social bonding and stress modulation [17]. Similarly, listening to 432 Hz tuned music, as opposed to the standard 440 Hz, has been shown to be more effective in reducing anxiety and stress among healthcare professionals [18].

Tempo is another critical factor influencing stress levels. Slow-tempo music (typically 60–80 beats per minute) has been consistently associated with reductions in heart rate, blood pressure, and cortisol levels, facilitating a greater sense of relaxation [4, 19]. In contrast, fast-tempo music may even induce mild physiological arousal or even increase stress in certain contexts. Akimoto [20] further confirm that slow-tempo, instrumental, and high-frequency music are among the most effective characteristics for stress reduction.

Although genre and musical features can provide a foundation for stress-reducing interventions, there is no universal 'best' type of music for stress relief. Stress-relief songs typically fall into two broad categories: soft, major-key compositions that promote tranquility and relaxation and energetic, minor-key pieces that help process stress through stimulation and emotional engagement [21]. Many effective stress-relief compositions are performed in the key of E, featuring moderate tempos and 4/4 time signatures, highlighting the role of rhythm and harmony in relaxation. These findings suggest that stress reduction through music is not a *one-size-fits-all* approach; while some individuals may benefit from soothing, slow-tempo music to induce relaxation, others may find relief through more dynamic and stimulating tracks that help them process emotions.

1.3.3 Music personalisation and stress reduction

Research increasingly suggests that personalisation is key to maximising music's stress-relieving potential. Music preference, listening context, and emotional state all play essential roles in how individuals experience stress reduction. For instance, some individuals may find energetic music helpful in boosting mood, while others may prefer slow, meditative tracks for relaxation [22]. Fully incorporating personal preferences into music therapy has been shown to significantly improve stress reduction outcomes [23].

These findings have significant implications for the development of personalized music recommender systems for stress reduction. Utilising machine learning algorithms and user data, recommender systems can curate music selections tailored to an individual's stress levels, personal

preferences, and activity context. This personalised approach has the potential to enhance the effectiveness of music as a stress-management for students, who, despite sharing a similar age group, can come from diverse cultural and musical backgrounds and experience stress from different sources.

1.4 Music recommender systems and stress reduction

One effective approach to personalising stress reduction through music is the use of music recommender systems. With the mass digitisation of audio content on streaming platforms, recommender systems have gained significant relevance, enabling users to navigate vast music collections efficiently and discover songs tailored to their preferences [24]. The ability of recommender systems to personalise music consumption utilising people's preferences and mood, could increase the effect of music on reducing the level of stress. The most commonly used methods for music recommender systems are collaborative filtering, content-based filtering, and a hybrid approach that integrates elements of both techniques.

1.4.1 Collaborative filtering

Collaborative filtering (CF) models recommend music based on user-item interaction data, capturing both explicit feedback (e.g., ratings) and implicit feedback (e.g., play history, skips) [25]. User-based CF methods identify users with similar listening patterns and suggest songs based on shared preferences, while item-based CF recommends songs similar to those a user has previously enjoyed.

Despite its widespread use, CF-based systems suffer from cold start problems, where recommendations lack relevance for new users due to insufficient interaction data [25]. Furthermore, traditional CF assumes that user preferences remain static over time [26]. However, stress is inherently dynamic, influenced by situational and emotional factors that fluctuate. As a result, a fully CF-based approach may fail to capture the evolving nature of stress-induced music preferences.

1.4.2 Content-based filtering

While CF methods largely focus on the similarities between users through user-item interactions, content-based filtering music recommender systems recommend songs based on the similarities between meta-data of the items the user has accessed before and that of other songs in the database. The item meta-data may include factual information such as "songwriter, singer, genre, style [...]" [27], high-level acoustic features such as instrument and genre, and low-level features such as rhythm, pitch, and spectral features (e.g. spectral flux and roll-off) [24, 28]. The focus on the target user provides this method with an advantage in personalisation. In an early work by Kim et al. [28], low-level features extracted from a user's music list were used to identify clusters within a user's preferences using a k-Means clustering algorithm. The clusters captured characteristic groups in the user's diverse range of preferred

music, namely in terms of genres. In the context of this research, a similar method may be applied to identify various groups of songs used by students for stress reduction.

1.4.3 Hybrid and context-based approaches

Hybrid systems combine the strengths of the previously mentioned approaches in order to mitigate the weaknesses of individual methods and improve recommendation. Apart from the three conventional techniques, context-based and mood-based techniques are emerging and relevant methods in music recommender systems research enabling further personalisation [29]. Context-based recommender systems incorporate contextual information regarding the user, item, and interactions. A study by Schedl et al. [30] concludes that user characteristics largely influence the performance of the music recommender system, thus highlighting the need to personalise recommendations by tailoring to the audience.

Emotion-based recommender systems go one step further by adapting recommendations in real time based on a user's current emotional state [29]. Previous works presenting music recommender systems for stress-reduction fall in this category. For instance, the recommender tool developed by Mustafa [31] applied face-recognition technology to infer the emotional state from the facial expression of the user, which replaced electroencephalogram (EEG) commonly used in similar tasks. Another approach by Shin et al. [10] utilised photoplethysmography (PPG) to assess stress-levels from heart beats. While these works have explored different techniques to measure stress in users, they have notably relied on pre-labeled song databases. As a result, there remains a significant gap for developing music recommender systems that are able to dynamically adapt to the desired stress-reducing strategy of the user that are captured by audio features.

1.4.4 Proposing a hybrid music recommender system for students

Building on these findings, this work will contribute a prototype of a hybrid music recommender system combining content-based filtering and collaborative filtering methods. The core of the algorithm will consist of a k-Means clustering algorithm, grouping student-curated songs for stress reduction based on extracted low-level audio features. This clustering approach not only groups similar music together but also provides insight into different students strategies for stress reduction, in regards to characteristic low-level audio features.

To further enhance user experience, the system will feature an interactive interface, allowing students to navigate between different music clusters based on their mood and situational needs. While the cold start problem presents initial challenges in implementing a fully functional CF-based system, the prototype will include a user feedback mechanism, enabling students to like and dislike recommended songs. Over time, this feedback will be incorporated into the system, refining recommendations by ranking songs within clusters based on the preferences of similar users.

By integrating machine learning techniques, real-time feedback, and personalized clustering, this approach aims to create a more adaptive and user-centered stress-reducing music recommender system, bridging the gap between existing theoretical frameworks and practical application.

2. RESEARCH QUESTION

Our research aims to address the main research question, guided by a few sub-questions that facilitate the exploration of key concepts and knowledge.

2.1 Main Research Question

Can a hybrid music recommender system be developed to support emotion-based stress coping for university students?

2.2 Sub-questions

1. What are key musical features (e.g., tempo, loudness, etc.) that contribute to stress reduction in university students?
2. Can we identify groups of songs used by university students for stress reduction by applying a k-Means clustering algorithm on acoustic features?
3. How effective is the music recommender system in providing students with personalised songs to support stress coping for university students?

3. METHODOLOGY

3.1 Methods

3.1.1 Exploratory Survey on Stress-Relief Songs

An exploratory survey was conducted to build a dataset of songs that students listened to when experiencing stress (Appendix D). The participants were asked to provide between one and five songs (song name and artist name) that they listened to during stressful moments. The survey was supplemented with basic demographic information and self-reported experiences of stress from the participants. The survey was distributed using a snowball sampling approach, whereby initial respondents — students from social networks — were encouraged to share the survey with their student peers. This method extended the survey's reach beyond the immediate circle of researchers. It ensured a broader and more diverse sample of students.

3.1.2 Prototype Testing

Usability testing was carried out to assess the usability and effectiveness of the designed stress relieving prototype. This usability testing was followed by a survey to capture participants' experiences during these tests (Appendix E). The assessment of the testing regarded the output and functionality of the prototype and not the visual aesthetics. The testing focused on four key areas: (1) usability and design, (2) functionality, (3) content, effectiveness and personalisation, and (4) overall user experience. The duration of the

usability testing was ten minutes. In order to maintain participant engagement and minimize fatigue. Ten minutes was enough time to yield accurate and meaningful feedback, because the primary user interacting - shuffling and liking songs- required approximately 30 seconds per song. This ten minute duration ensured that the participants engaged with approximately 20 songs. Which provided a sufficient sample size for meaningful insights. The detailed testing protocol and consent form used in this study can be found in appendix B and C.

3.1.3 Dataset

The studies dataset was built with stress-relieving songs from multiple sources. The first dataset consisted of a total of 38 songs extracted from a student-curated stress relief playlist compiled by students in the 2024-2025 iteration of the Sound and Music Technology course at Utrecht University (hereafter, referred to as "SMT playlist"). The second dataset consisted of 194 additional songs collected through the exploratory survey. This brought the total to 232 stress relieving songs in the dataset. The criteria for the dataset were a minimum of 200 songs. These songs were refined before being added to the dataset. Some students entered entire albums, genres or songs that exceeded ten minutes in length. These survey inputs did not meet the criteria of song artist and song title. The songs were grouped into clusters to analyze patterns in the dataset. Details are described in section 3.3.

3.2 Participants

3.2.1 Survey Participants

The exploratory survey gathered responses from 42 participants. The inclusion criteria for participating in this part of the research were broad. The participants were only required to be students. The sample size was determined based on the number of useful stress-relieving songs for the dataset (232 songs). The average age of the participants was 21 years ($M=21$, $SD=3.62$). The gender distribution was: 57.1% male ($n = 24$), 33.3% female ($n = 11$), 7.1% non-binary ($n = 3$), and 2.4% other gender-conforming identifies ($n = 1$). 92.9% of the participants were currently enrolled in post-secondary education. The other 7.1% ($n=3$) participants had just graduated. In terms of stress experiences, 87.5% ($n = 35$) reported experiencing stress due to both academic and other factors in their life. Whereas 10% ($n = 4$) attributed their stress solely to non-academic factors, and 2.5% ($n = 1$) reported experiencing stress only due to academic reasons. None of the participants indicated having no stress at all. Music was commonly used as a coping mechanism. 85% ($n = 34$) participants said that they listened to music when feeling stressed. The other 15% ($n = 6$) of the participants did not listen to music to reduce stress.

3.2.2 Prototype Testing Participants

17 students participated in the usability testing. The age distribution ranged from 18 to 27 years old. Most participants were 21 years old ($n = 5$, $SD = 2.44$). The gender

distribution was: 58.8% female ($n = 10$), 35.5% male ($n = 6$) and 5.9% was non-binary ($n = 1$). In terms of education, 88.2% were enrolled in post-secondary education ($n = 15$). Whereas 11.8% of the participants were in secondary education ($n = 2$). Eligibility criteria required that the participants were students and had experienced stress. This ensured that the participants had relevant lived experiences to provide useful feedback. The emotional stage of the participants during the prototype testing was not part of the criteria. The participants were asked to select/like songs they would have listened to when feeling stressed.

3.3 Audio Feature Extraction and Clustering Algorithm

To address sub-questions (1) and (2), audio features were extracted and clustered using the k-Mean clustering method. Subsequently, these clusters were analysed by combining quantitative and qualitative findings to identify groups of songs. The aim of this cluster analysis is to gain insight into the key audio features that differentiate various student-specific strategies of music consumption for stress reduction, and to build a foundation for the subsequent development of the hybrid recommender system algorithm.

A total of 38 songs were downloaded in a mp3 format from the SMT playlist. The audio files were used to extract audio features using *librosa*, a Python-based library for audio and music processing. The following sections will outline the following stages: audio feature extraction, feature-selection, and k-Means clustering. In the results section, the resulting clusters will be analysed through the examination of audio feature values and the comparison of clusters to student-reported explanations of stress-reducing purpose and song characteristics.

3.3.1 Audio Feature Extraction

Using the *librosa* Python package, all mp3 files were converted to audio time series before audio feature extraction. The audio time series for each song was separated into harmonic and percussive components to obtain the chromagram and tempo, respectively. The chromagram was utilized for the estimation of key and mode using the Krumhansl-Schumakler key finding algorithm [32]. A total of nine audio features were extracted: *Key*, *mode*, *tempo* (*beats-per-minute*), *root-mean-squared (RMS)*, and several spectral features encompassing *centroid*, *bandwidth*, *contrast*, *flatness*, and *roll-off*.

For spectral features with a time component, the values were aggregated (mean) over the time dimension due to constriction of the k-Means clustering algorithm and computational limitations. However, this aggregation approach may have several implications for the efficacy of audio features in the comprehensive representation of songs: this will be further discussed in section 5.2.

3.3.2 Pre-processing and Feature Selection

Continuous audio features (*tempo*, *RMS*, *centroid*, *bandwidth*, *contrast*, *flatness*, and *rolloff*) were selected for k-Means clustering. These features were standardized to

limit bias in the clustering algorithm relying on Euclidean distances: the audio features were transformed to have a mean of zero and a standard deviation of one. Furthermore, pairwise feature correlation between features were calculated to identify highly correlated features. The correlations revealed that *spectral centroid* is highly correlated with *spectral bandwidth and rolloff*. In order to prevent the risk of multi-collinearity, *bandwidth* and *rolloff* were removed from subsequent processing.

In an effort to further reduce dimensionality, a principal component analysis was attempted through an examination of explained variance by the number of components. An explained variance of 95% (a common threshold) was surpassed at five components, thus, justifying the need for the five remaining features (*tempo*, *rms*, *centroid*, *contrast*, *flatness*) without dimensionality reduction.

3.3.3 k-Means Clustering

An unsupervised k-means clustering, initialized by a greedy k-means algorithm, was performed on five audio features. The optimal number of clusters was identified using the ‘elbow method’: The within-cluster sum of squares (WCSS) values were plotted against an increasing number of clusters. The optimal k-value (i.e. the ‘elbow’) was identified at 4 clusters. The four clusters were quantitatively and qualitatively analysed to infer various student strategies in stress reduction through music consumption.

3.4 Prototype Design

The prototype is designed to explore the efficacy of clustering-based music recommendation systems in tailoring song selections to user preferences. As described in the methodology, it integrates unsupervised learning techniques with an interactive user interface to enhance personalized music discovery.

This prototype builds upon the method described in the clustering section. The clusters are central to the prototype, employing k-Means to group songs based on the aforementioned audio features. The audio features are standardized and weighted to improve clustering fidelity. A weight was added to adjust more bias/focus to tempo. As can be seen in figure 1, 232 songs had been compiled through the SMT playlist and the exploratory survey. These songs were then placed into eight different clusters. Eight was chosen as a starting point to account for the increased number of songs, and will be a point of optimization and testing for future iteration.

The recommendation engine utilizes these clusters to refine song selection dynamically. Upon initialization, users specify preferences related to tempo and energy, which guide the selection of an initial cluster. As users interact with the system — expressing preferences for faster or slower tempos, greater or lesser intensity, or shifts in musical mode — the prototype dynamically adjusts recommendations by transitioning between clusters in a structured manner, preventing repetitive recommendations and

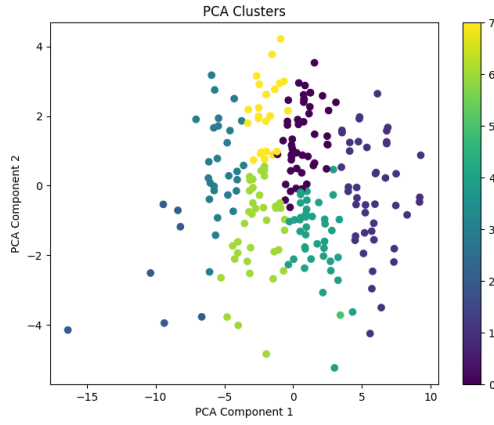


Figure 1. Distribution of 232 songs across 8 clusters based on their audio features.

improving diversity in song selection. Furthermore, an additional feature was added to enable the users to "like" songs that they encounter. The list of liked songs can be exported by the user to aid music discovery, and by the developers to gather user-item interaction data for future integration of the collaborative filtering element.

A basic graphical user interface (GUI) which is shown in Figure 2, was implemented to facilitate user interaction, enabling real-time adjustments to preferences and visualization of cluster transitions. User engagement metrics, including the frequency of preference modifications and transitions between clusters, are logged for analysis. Evaluation of the prototype focuses on the effectiveness of cluster-based recommendations in capturing user preferences, the responsiveness of the system, and overall user satisfaction. This approach enables systematic refinement of the clustering methodology and its practical application in music recommendation systems. Further explanations of the prototype is outlined in Appendix F, supported by a demonstration video in the supplementary files of the project.

4. RESULTS

4.1 Feature Extraction and Cluster Analysis Results

4.1.1 Preliminary Feature Analysis

The distribution of estimated tonics is shown in Figure 3. The tonics A, D, F, G, and B represent the majority of the songs. Interestingly, E was only detected in 7.7% of songs, contradicting the results presented in the literature review, suggesting that it is commonly used in one of the two song categories for stress relief [21]. However, the frequency distribution of modes in the student-curated stress relief playlist reveals that major and minor keys are represented approximately in equal degrees, with a slight domination of major keys (55%). This finding aligns with the categories identified by Adiasto [21], which suggest one category consisting of major-key tunes and another consisting of minor-key tunes.

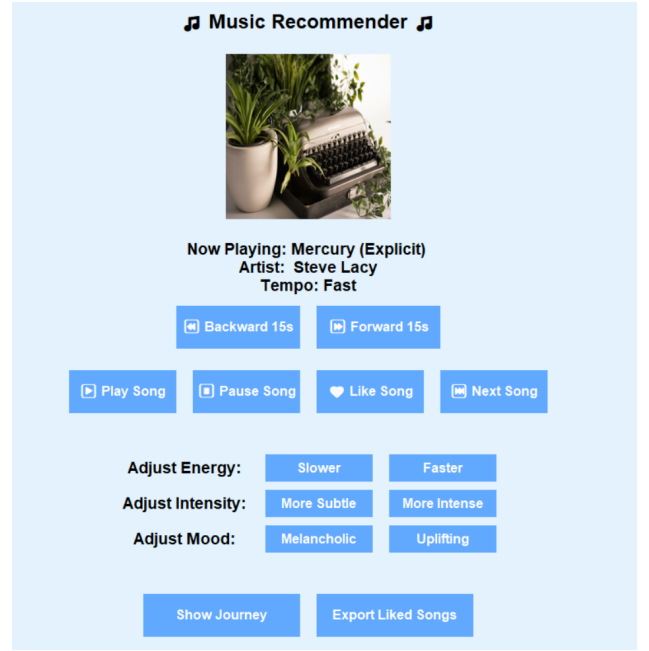


Figure 2. The GUI of prototype

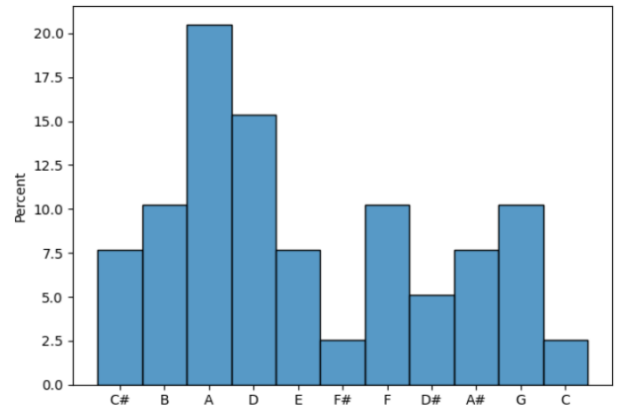


Figure 3. Distribution of estimated tonics of songs in the SMT playlist

Furthermore, the extracted tempo of the songs were approximately normally distributed ranging from 60-180 beats per minute, with a mean value of 120.6. While previous works find that music with slow tempo (60-80 beats per minute) are effective for stress reduction [4, 19, 20], less than 5% of the songs in the student-curated playlist were found to be in this tempo range. These findings suggest that the music consumption patterns of university students for stress relief may differ from findings in the general population.

4.1.2 Cluster Analysis

The k-Means clustering algorithm has identified four groups of songs in the student-curated playlist. The number of songs classified in each cluster and its median feature values are summarised in Appendix A.

A heatmap of standardized median feature values (Figure 4) illuminates the characteristic feature values of each

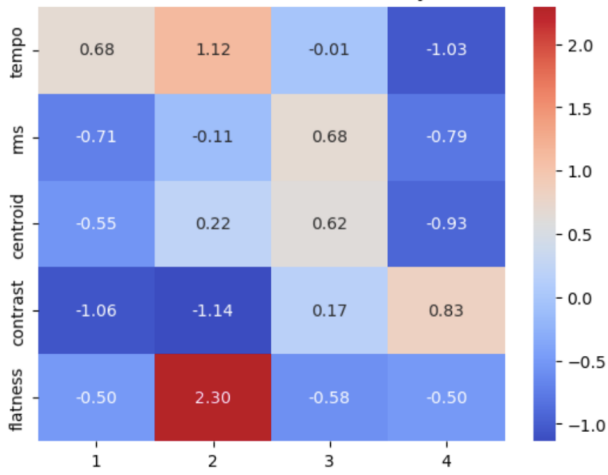


Figure 4. Standardised median feature values of clusters obtained by applying k-Means clustering on the SMT playlist

cluster. Clusters 1 and 4 both exhibit relatively lower values in *RMS*, *centroid*, and *flatness*, indicating softer, lower-toned, and tonal qualities of songs in the clusters. However, the median *tempo* values differentiate the two clusters: Cluster 1 exhibits a moderately high *tempo* (0.68) as opposed to Cluster 4’s characteristic low *tempo* (-1.03). Additionally, Cluster 2 is characterised with high *tempo* and *spectral flatness* values, implying a more energetic and noise-like nature. Cluster 3 is characterised by relatively higher *RMS* and *centroid* values implying a louder, high-toned nature of songs in this cluster.

These clusters were qualitatively compared to student-written text, in which students described why each song suited their stress-reducing purpose (*"Why this song/piece?"*). These texts were used to validate and contextualise the quantitative findings resulting from the cluster analysis.

- In student explanations for songs in Cluster 1, words such as “relaxing”, “mellow”, and “meditative” were often used. These explanations align with the low *RMS*, *centroid*, and *flatness* values of the cluster. Students also tend to mention the instruments in the piece, which may indicate the tendency of instrumental songs.
- Words such as “upbeat”, “motivating”, and “hopeful” were frequently used as descriptors for pieces in Cluster 2. These explanations align with the high median *tempo* of the cluster. Additionally, 3 out of 5 students reported nostalgia or childhood associations with the song.
- Frequent mentions of “energy”, “hope”, and “strength” were seen in descriptions corresponding to songs in Cluster 3, similar to Cluster 2. Students often also mention the rhythm and beats of the song. Many songs in this cluster were electronic or otherwise had clear percussive elements, which align with the high *tempo* and *flatness* values of the cluster.

- Student explanations for Cluster 4 often mentioned the pieces to be “soothing” and “peaceful”. Almost all of the students had reported a sense of calm associated with the song’s listening experience. The low *RMS*, and *centroid* values, and especially the low *tempo* seem to align with these reports.

The subjective reports of students demonstrate similarities between Clusters 1 and 4, aligning with the similarities in their characteristic audio feature values, namely *RMS*, *centroid*, and *flatness*. The songs in these clusters often offer a calming and relaxing experience, aligning with the inferred qualities of the audio feature values. However, Cluster 1 seems to have a stronger focus on instruments compared to cluster 4 based on student reports, which arose from trends in student explanations.

While the similarity between clusters 2 and 3 were not immediately apparent through the comparison of audio feature values, the student reports demonstrate the tendency for more energetic and hopeful music. Cluster 2 seems to have correlations with the feeling of nostalgia, and therefore, may focus on older pieces. On the other hand, the electronic nature of cluster 3 could imply the preference for newer musical pieces.

4.1.3 Implications

From the established similarities between Clusters 1 and 4, and Clusters 2 and 3, two student strategies in stress-reduction can potentially be deduced. The cluster statistics and student explanations imply that students generally choose between calm or energetic songs for stress reduction. While previous works have demonstrated the effectiveness of slow to moderate-paced music as opposed to high-paced music in stress reduction, the clustering analysis has shown that some students actively utilise high tempo songs to alleviate stress. This variability further highlights the need for the personalisation of music recommendation in order to effectively cater to individual stress-coping needs.

4.2 Prototype and User Testing Results

To evaluate the performance of our prototype in reducing stress (*sub-question (3)*), we designed a user test. During each test, the participant learned about our research objective and how the prototype works. After a participant freely experienced the prototype for 10 minutes, they were be asked to complete a questionnaire to evaluate our prototype.

4.2.1 Quantitative test results

We used a 1-5 Likert scale to quantify users’ experience with the prototype on 3 metrics. Table 1 shows the average score for each metric.

- *Acceptance of design idea.*
Except for 1 (5.9%) participant who was neutral about our design concept, all other participants agreed with our concept, especially 6 (35.3%) participants of them totally agreed with our design idea.

- *Ease of use*
During the experience, no one felt that the prototype was not easy to use. 6 (35.3%) participants were neutral about the ease of use, while 6 (35.3%) participants felt the prototype was easy to use and 5 (29.4%) found it extremely easy.
- *Satisfaction with the function of adjusting preferences*
As for the function of adjusting preferences, no one thought the function did not work properly at all. Unfortunately, there are still 2 (11.8%) participants who thought this function was not that useful. Besides these participants, 6 (35.3%) participants were neutral about the functionality, while 2 (11.8%) participants felt the function works to a certain extent and 7 (41.2%) thought it works perfectly.

Metrics	Score
Acceptance of design idea	4.29 / 5
Ease of use	3.94 / 5
Satisfaction with adjusting preferences	3.82 / 5

Table 1. Mean of the scores for each metric

4.2.2 Qualitative test results

Qualitative questions included participants' experienced technical issues, the relevance of music choices, and the accuracy of song recommendations.

- *Technical issues*
There are 3 (17.6%) participants who reflected they met some problems including a song may appear repeatedly with different background images, and the buttons for adjusting the speed of songs and switching to the next song do not work well.
- *Stress relief and preferences match*
When participants were asked whether the recommended music relieved stress and whether the recommended music matched their preferences, 14 (82.4%) of them thought the prototype provided the user with appropriate music choices for stress relief. The same number of participants said the recommended songs met their preferences.
- *Usage willingness and recommendation accuracy*
When participants were asked whether they would use the prototype in the future and whether the music recommendations met their expectations, 14 participants reported positively. 12 (70.6%) participants would like to use the prototype to relieve stress while the same number of participants thought the recommended song after answering the questions in the very beginning met their expectations.

4.2.3 Suggestions feedback

The final open questions accommodated user feedback about the design concept, user interface design, and preference adjustment function of this prototype. These functions of these features were perceived positively by some

participants. In addition, some participants believed that the prototype provided a wide variety of song types and high quality, which met their preferences.

Participants also gave some suggestions for improvement. Some participants hoped the prototype could add more categories of songs and make more detailed classifications. Additionally, there can be more factors when adjusting preferences to make the function more flexible. The prototype could have a more detailed introduction to each button's function, as one participant expressed he was confused about the intensity-adjusting feature. Otherwise, some participants hoped the background image of the song could be replaced with the cover of the album to which it belongs, and favorite songs could be exported to external applications.

5. DISCUSSION

This study demonstrates the potential for a hybrid music recommender system to support emotion-based stress coping for university students. A comprehensive literature review encompassing various topics — stress among students, the relationship between music and stress, and music recommender systems — has underscored the purpose of developing a personalised, user-centric, and adaptive music recommender system for university students. Notably, despite 85% of students reporting that they listen to music when stressed (section 3.2.1), prior research has largely overlooked student-specific music-based coping mechanisms. Our study contributes to the field in three key ways: (1) corpus analysis of a student-curated playlist for stress-reduction, (2) cluster analysis based on extracted audio features, and (3) prototype development and evaluation of a hybrid music recommender system for student stress reduction.

5.1 Interpretation of Results

5.1.1 What are key musical features that contribute to stress reduction in university students?

Eight audio features were extracted from the audio files of a playlist of 38 songs, capturing tonality, rhythm, and other low-level spectral characteristics. Five features — *tempo*, *root mean square (RMS)*, *spectral centroid*, *contrast*, and *flatness*— were selected for k-Means clustering after excluding categorical features and removing highly correlated spectral features to mitigate multicollinearity. An examination of the explained variance confirmed that these five features accounted for 95% of the variance between clusters, establishing them as key differentiators of stress-relief music among students.

The distributions of extracted audio features were analysed and compared to prior research on musical features and stress reduction. In general, the variability in audio feature values reflected the general lack of consensus on stress-relieving musical features. While the mode distribution is aligned with findings by Adiasto [21], other features deviated from findings in previous research. For instance, prior studies suggest that slow-tempo music (60-80bpm)

is most effective for stress reduction, while faster tempos may elevate stress levels. [4, 8, 20]. However, the student-curated SMT playlist had a mean tempo of 120 bpm, with nearly half of the songs falling within the 120–180 bpm range. These results may suggest that student-preferred stress-relief music may differ from those identified in studies on the general population, presenting an interesting avenue for further research focusing specifically on students.

5.1.2 Can we identify groups of songs used by university students for stress reduction by applying a k-Means clustering algorithm on acoustic features?

The cluster analysis on the four groups of songs classified by the k-Means algorithm highlighted two broad categories of music used for stress reduction by students. One category included songs which were described as “calming” and “relaxing”, while the other category encompassed songs that were described as “energetic” and “hopeful”. These findings align with Adiasto’s categorisation of music for stress relief [21], which distinguishes between tranquil music that promotes tranquility and relaxation, and another category including energetic pieces that help process through stimulation and emotional engagement. While similarities between clusters were inferred from the subjective description provided by students, notable differences in audio feature values were also observed within clusters belonging to the same category. These variations may suggest that students have more nuanced and fine-grained preferences within the broad categories, which can be inferred through differences in audio feature values.

5.1.3 How effective is the music recommender system in providing students with personalised songs to support stress coping for university students?

The developed prototype integrated the k-Means clustering approach, which allowed users to dynamically adjust recommendations by expressing preferences in tempo, energy, and mode through buttons. This study contributes a novel user-centric approach for music recommendation systems, contrasting previously proposed systems that have focused on physiological cues of stress to adapt recommendations [10, 31]. By shifting the agency of recommendations towards students, the proposed recommender system moves away from the passive, one-fits-all approach of current systems. Instead, students are able to actively tailor recommendations based on their musical preferences and mood.

The user testing provided valuable insights into the prototype’s effectiveness and usability. The acceptance of the design was rated highly on average (4.29/5). Furthermore, 82.4% of participants reported that the prototype was able to meet their musical preferences, and 70.5% of participants reported that they would like to use the prototype to relieve stress. These results suggest the effectiveness of the proposed system design in aiding personalised stress-reduction in students. However, areas for improvement were identified, particularly in ease of use (rated 3.94/5) and the functionality of the adjustment buttons (3.82/5).

5.2 Challenges and Limitations

The user testing has demonstrated the promising potential of the proposed hybrid recommender system, however, has also illuminated several limitations of the prototype.

5.2.1 Functionality of the feature-adjustment buttons

One of the areas of improvement identified by participants is the functionality of the feature adjustment buttons. In particular, the tempo adjustment buttons were reported to be less functional than the other buttons. These errors were the most apparent in songs with little or no percussive instrumentation, such as classical piano pieces. The initial prototype applied the tempo estimation algorithm after the harmonic-percussive separation of the audio signals. After encountering these errors, tempo estimations were run on audio signals prior to this audio signal separation. However, the observed changes in the estimated tempo were minimal (< 0.2 bpm) implying inherent limitations in *librosa*’s tempo estimation function. In future works, more advanced techniques such as Inner Metric Analysis can be explored as a replacement to this function.

5.2.2 Balancing Dimensionality Reduction and Temporal Information in Spectral Audio Features

Furthermore, as outlined in section 3.3.1, spectral features were aggregated for dimensionality reduction, compromising temporal information. The motivation behind the aggregation was to cater to the nature of k-Means clustering algorithm — the heart of the prototype — which requires audio features expressed as fixed-length vectors. The dimensionality reduction also improves the scalability of the system, as the amount of data is compressed. While dimensionality reduction improves scalability, it also limits the system’s capacity to capture changes in audio features over time. To counteract this trade-off without significantly compromising scalability, additional metrics such as standard deviation can be used to supplement the aggregated values.

5.2.3 Dataset limitations and user demographics

Another suggestion for improvement identified by a participant is the expansion of the current song database with a more diverse set of songs to accommodate a larger demographic of music tastes. This highlights the limitation of our corpus curation process, which relied on the social network of the researchers (i.e. snowball sampling), potentially limiting the full representation of the diverse range of stress-relief songs listened to by students. User testing can also benefit from a larger and more diverse sample of students to better probe the potential of the hybrid recommender system in wider, real-life contexts.

5.2.4 The absence of fully integrate collaborative filtering

Finally, the collaborative filtering element has yet to be integrated into the current prototype for a fully functioning hybrid recommender system. The integration of collaborative filtering was out of the scope of this study due to the lack of user-item interaction data. Commonly used

datasets in the research domain of music recommender systems contain a vast amount of user-interaction data, needed to build high-quality collaborative filtering models. For example, the LFM-2b dataset encompasses data from 120,000 users, 50 million tracks, and 5 million artists [30]. Nevertheless, through the user testing of the current prototype, implicit and explicit user-item interaction data was collected by logging the users' visited clusters and liked songs. This data can be amassed over time and utilised to rank recommendations, by integrating user-based and/or item-based collaborative filtering algorithms.

6. CONCLUSION AND FUTURE WORK

In this study, we presented a prototype of a hybrid music recommender system designed to support healthy emotion-based stress-coping in students through music. The user testing suggests that the system's user-centric, dynamic, and personalised design is effective for stress reduction, indicating a promising potential for its implementation. A review of the existing literature revealed significant knowledge gaps regarding students' use of music for stress relief and the role of personalisation in music recommendations.

The audio feature extraction and cluster analysis of a student-curated stress-reduction playlist contributes an analysis of audio features in stress-reducing music used by students. The findings suggest that characteristics in student-curated songs differ from those that are found to be effective for stress reduction in the general population. The k-Means clustering approach revealed two broad categories of stress-relieving music used by students: calming and energetic. The findings and algorithms in the cluster analysis were inherited by our prototype, emphasizing a user-centered approach to cater to individual preferences. The success of this design was validated through the user testing of the prototype with students, indicating promising potential for this system.

Future research should focus on integrating collaborative filtering techniques, to develop a fully functional hybrid recommender system. Additional improvements include refining the audio feature extraction process, expanding and diversifying the song database, as well as increasing the number and diversity of the participants in user testing. To further advance the system's performance, state-of-the-art techniques such as deep clustering can be explored to capture nuanced qualities of user preferences and musical pieces, thereby improving both personalisation and clustering abilities.

7. REFERENCES

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8. APPENDIX

A. MEDIAN CLUSTER VALUES

Cluster	Nr. Songs	Tempo	RMS	Centroid	Contrast	Flatness
1	9	139.7	0.1050	1689	13.77	5.520×10^{-4}
2	5	152.0	0.1490	2188	13.52	9.950×10^{-3}
3	16	120.3	0.2060	2442	17.73	2.900×10^{-4}
4	8	91.48	0.1000	1440	19.84	5.590×10^{-4}

Table 2. (Median) cluster statistics rounded to 4 significant figures

B. PROTOCOL USABILITY TESTING PROTOTYPE

Introduction to the study

The testing begins with welcoming and thanking the participants for their willingness to contribute to this

study. The purpose of the research will be explained to the participants. This is followed by an explanation that the goal is to evaluate a prototype music platform. Which is designed to help students discover songs that can ease their stress. The music gathered in the platform is created through students, for students. The facilitator will clarify that the evaluation focus is on the usability of the platform and the accuracy of its output in recommending stress-relief songs. The participants are asked to engage naturally with the prototype and find songs they would listen to in order to reduce stress. The significance of their participation, in improving the prototype, will be highlighted.

Consent Form

The participants are provided with consent form before proceeding with their participation. The consent form outlines key ethical considerations:

- *Voluntary participation:*
The participants are able to withdraw from the research at any time without any consequences.
- *Anonymity and confidentiality:*
The responses from the participants will remain confidential and used solely for the purpose of this research.
- *Use of data:*
The data that is collected during the testing will only be used for research purposes. It will not be shared with third parties for commercial purposes.

Prototype Explanation

The facilitator will introduce the prototype and explain the testing procedure:

- *Task:*
The participants will have 10 minutes to explore the platform and shuffle through the songs in order to
- *No need to feel stressed:*
The test does not require the participants to feel stressed when exploring the prototype. They should rather select songs that they would listen to in stressful situations.
- *Post-test survey:*
The participants are asked to complete a short survey to share their opinions and experiences after using the prototype.
- *Use of data:*
The selected/liked songs and the clusters the participants visited, will be used for this research purposes.

C. CONSENT FORM

Research Participant Information Sheet: The Rhythms of Relief: Designing a Stress-Relieving Music Recommender System for University Students

Daniel Bourdon, Linka Mitome, Tim Kramer, Louisa Falch,
Shirin Ghafarijafari and Binxu Jiang
Utrecht University, January 22, 2025

What is the purpose of this study

This research focuses on designing a personalized music recommendation system. Which is aimed at reducing stress among university students. The study combines insights from music therapy, cognitive psychology, and technology to identify key musical features and preferences for stress relief. A prototype was created with the help of audio data analysis and machine learning. This prototype is developed to generate a personalized music stress-relieving playlist. This research provides an answer to the research question: ‘How can a personalised music recommendation system be designed to optimise stress reduction for university students?’ And its sub-questions: ‘What are the key musical features (e.g., tempo, key, energy) that contribute to stress reduction in students?’ and ‘How do individual differences (e.g., gender, stress levels, musical preferences) impact the effectiveness of music for stress relief?’.

What will I do if I choose to be in this study?

The role of the participant in this part of the research project is to interact with and evaluate the prototype that was created.

How long will I be in this study?

The participation in this study will be around 20 minutes. The researcher(s) will start with explaining the prototype to the participant. When the participants understand how the prototype works, the active part of the research will start. The participants have 10 minutes to move around in the prototype. Selecting stress relieving songs they like and skipping songs they dislike. After this is done, the participants are asked to fill in a short survey. The survey will ask the participant questions about their opinion of the prototype, the personalisation part of the prototype and its usability.

What are the risks or discomforts?

Participants may find that some recommended songs do not align with their personal preferences. This could cause potential frustration or disappointment. Furthermore, the research can take some time, which can be inconvenient for the participants. Lastly, sharing preferences or opinions related to music and stress might feel uncomfortable for the participants, even though all the data will be anonymized.

What are the advantages?

The participants will be provided with their own list of selected songs they have chosen from the dataset. These potentially newly discovered songs may help them in reducing stress. The participants will also help with advancing the development of mental health solutions tailored to university students.

What are my rights if I take part in this study?

Participation in this study is completely voluntary and participants can choose to withdraw from the study at any time, without providing an explanation or facing any consequences. All data collected during the study will be anonymized and handled confidentially. Personal information will not be used or shared for any purpose outside of this study. Furthermore, participants can request to see the data collected about them and can ask questions about the research and their involvement. The researchers are obligated to provide the participant with clear answers.

Who can I contact if I have question about this study?

In case you have any questions, comments or concerns about this research, please feel free to contact one of the researchers via email: l.k.j.falch@students.uu.nl and l.mitome@students.uu.nl.

D. EXPLORATORY SURVEY - MUSIC FOR STRESS RELIEF

1. What is your current age?
2. How do you identify?
 - Man
 - Woman
 - Non-binary
 - Else...
3. What level of education are you currently enrolled in?
 - Secondary education (e.g., high school, secondary school, or equivalent)
 - Post-secondary education (e.g., university, college, technical or vocational training)
 - not currently enrolled in any educational program
4. Do you sometimes experience stress (physical and/or mental)?
 - Yes, but not related to my education
 - Yes, but only related to my education
 - Yes, both related to my education and other factors in my life
 - No
 - Else...
5. Do you sometimes listen to music to regulate stress?
This includes both academic and non-academic related stress
 - Yes
 - No

6. What are musical pieces or songs you choose to listen to in order to regulate stress?

Please share at least (1) the title and artist of the song, and (2) a brief description on why you picked that specific piece. You can also include a Spotify and/or YouTube link. Feel free to share at least one, but up to five musical pieces or songs.

- Open question...

E. SURVEY USABILITY TESTING - MUSIC FOR STRESS RELIEF

1. What is your current age?
2. How do you identify?
 - Man
 - Woman
 - Non-binary
 - Else...
3. What level of education are you currently enrolled in?
 - Secondary education (e.g., high school, secondary school, or equivalent)
 - Post-secondary education (e.g., university, college, technical or vocational training)
 - not currently enrolled in any educational program
4. On a scale from 1-5, how easy was it to navigate the prototype?
Very difficult ○ ○ ○ ○ Very easy
5. On a scale from 1-5, What did you think about the overall design 'idea' of the platform?
Very Poor ○ ○ ○ ○ Very excellent
6. On a scale from 1-5, how well did the control buttons (energy, intensity, mood) work in your personal experience?
Not at all ○ ○ ○ ○ Very well
7. Did you experience any technical issues?
 - Yes
 - No
8. If you answered 'yes' in the previous question, what kind of problems did you experience?
9. Do you think that the platform provides the user with appropriate music choices for stress relief?
 - Yes
 - No
10. Would you use this app to listen and find music to help you relieve stress?

- Yes
 - No
11. Do you feel like the prototype recommended the first song according to the question (i.e., "Would you like the music to be fast or slow?") you answered in the beginning?
- Yes
 - No
12. Could you explain why you gave this answer?
13. Do you think the programme was able to recommend songs that match your preferences?
- Yes
 - No
14. What did you like most about the prototype?
15. Do you have any feedback in improving the prototype?

F. PROTOTYPE EXPLANATION

The prototype is designed to provide a calming and intuitive music experience. The light blue colour scheme was chosen for its soothing effect. The users have access to several interactive features to personalize their listening experience while engaging with the prototype.

They can navigate through songs using the "Backward 15s" and "Forward 15s". These buttons let the user rewind or skip small sections of the song. It is also possible to put the song on pause and resume when wanted. When a user finds a song they would listen to when stressed, they can press "Like song" to save it. If the user dislikes a song, it is possible to skip directly to the next track with "next song". The user has the possibility to further tailor their experience. Users can switch songs based on energy level, by choosing either a slower or faster track. They also have the possibility to adjust the intensity of the music. The user can opt for either a more intense or subtle sound. The mood of the song can also be influenced. It is possible to skip to a more "melancholic" or "uplifting" song.

The prototype also has the option to visual represent the users listening journey. The user can click "show journey" to see how he navigated through the clusters. Finally, all liked songs are collected in a list. Which is visualized by the button "export liked songs".